

1. A *special-machine* is exactly like a one-tape Turing machine, except that the input string $x \in \Sigma^*$ is enclosed in left and right endmarkers, $\vdash$ and $\dashv$, which can not be overwritten, and the machine is constrained never to move left of the $\vdash$ nor right of the $\dashv$. It may read and write all it wants between the endmarkers. Let

$$A_{\text{special}} := \{ \langle M, w \rangle \mid M \text{ is a special-machine and accepts } w \}.$$

Show that A_{special} is decidable.

(9 marks)

2. Show that the language $\{ \langle G \rangle \mid G \text{ is a CFG and } 1^* \cap L(G) \neq \emptyset \}$ is decidable.

(9 marks)

3. Let $S := \{ \langle M \rangle \mid M \text{ is a DFA that accepts } \text{rev}(x) \text{ whenever it accepts } x \}$. Here $\text{rev}(x)$ is the reverse of the string x . Show that S is decidable.

(9 marks)

4. Let $\text{Int} := \{ \langle M_1, M_2 \rangle \mid M_1, M_2 \text{ are TMs such that } L(M_1) \cap L(M_2) \neq \emptyset \}$. Show that Int is undecidable.

(11 marks)

5. Let $T := \{ \langle M \rangle \mid M \text{ is a TM that halts on all input strings of length } \leq 100 \}$. Show that T is undecidable.

(11 marks)

6. Show that the language $\text{All} := \{ \langle M \rangle \mid M \text{ is a TM and } L(M) = \Sigma^* \}$ is not recursively enumerable.

(12 marks)

7. Show that the language $\text{Unit} := \{ \langle M \rangle \mid M \text{ is a TM and } |L(M)| = 1 \}$ is not recursively enumerable.

(12 marks)

8. Let NAESAT be the collection of satisfiable 3-CNF formulas Φ with a satisfying assignment that makes *at least one literal true* and *at least one literal false* in each clause of Φ . Show that the language NAESAT is NP-Complete.

(12 marks)

9. A *cut* in an undirected graph $G = (V, E)$ is a partition of the vertex set V into two disjoint subsets S and $V \setminus S$. The *size* of a cut is the number of edges that have one endpoint in S and the other in $V \setminus S$. Let

$$\text{MaxCut} := \{ \langle G, k \rangle \mid G \text{ has a cut of size at least } k \}.$$

Show that MaxCut is NP-Complete.

(15 marks)